


Karunya INSTITUTE OF TECHNOLOGY AND SCIENCES

(Declared as Deemed to be University under Sec.3 of the UGC Act, 1956)

 MoE, UGC & AICTE Approved; NAAC Accredited A++
 Karunya Nagar, Coimbatore - 641 114, Tamil Nadu, India.

School of Computer Science and Technology & Digital Sciences

(Division of CSE, AIML, DSCS, DS)

Part – A

MICRO-PROJECT PROPOSAL

I. Details of the Students:

S.No	Name of the Student	Register Number
1	SANTHOSA PRIYAN K A	URK25CS7100
2	KALAISELVAN R P	URK25CS7047
3	NISTCHAL MARCHUS Y	URK25CS7112
4	DHANUSH DHARMENDRA	URK25CS7086

II. Details of the Course related to the Micro-Project:

Course Title:	Venture Ideation and Entrepreneurship Practice
Course Code:	25MS204
Course Faculty:	B. Vidhya
Academic Year:	2025-2026
Semester:	II Even Sem
Year & Batch:	I year – AIML Batch No 3

III. Macro-Level and Meso-Level Sub-Tasks (COs) related to this Micro-Project:

Macro-Level Competency:	<u>Evaluate</u> the performance of electrical circuits, machines and systems	
Course Outcomes:	CO1	<u>Identify</u> entrepreneurial traits and match them with suitable business opportunities.
	CO2	<u>Validate</u> feasible business ideas using alternative entrepreneurial methods.
	CO3	<u>Formulate</u> a startup proposal for the chosen idea through entrepreneurial support systems.
	CO4	<u>Design</u> a suitable business model and formulate financial plans for the selected product/service.
	CO5	<u>Develop</u> a pitch deck of the identified product/service and its management strategies for ROI.

Title of the Micro-Project

1.0 **Rationale** *(Students should elaborate on the importance/significance of doing this project, in about 200 to 400 words.)*

Have you ever looked at your electricity bill and wondered, "Where did all these units go?" You're not alone.

We surveyed 54 households and found that **83% of people have no idea which appliance consumes the most power** in their home. Is it the AC running all night? The old fridge? The morning geyser? Most of us just pay the bill and move on.

The problem is real. An average Indian family spends ₹3,000-₹8,000 on electricity every month, and bills keep increasing 10-15% yearly. But here's the good news - our research shows that with simple awareness, households can **save 15-25%** on their bills. That's ₹9,000-₹15,000 back in your pocket every year.

UrjaSaathi (meaning "Energy Companion") is our solution. It's an AI-powered app that:

- Shows you exactly which appliance is eating how much power
- Gives personalized tips to save money
- Helps you apply for government solar subsidies in 15 minutes
- Speaks your language - Hinglish, Hindi, or English

Why does this matter? Because existing solutions are either too expensive or not built for Indian homes. International products cost ₹15,000+. Discom apps only show monthly totals. Smart plugs need ₹1,200 per appliance.

We're building something different - affordable, Indian-first, and actually useful.

From a business perspective, India has 250 million households and the smart energy market is growing at 18% annually. With 78% of our survey respondents saying they'd use UrjaSaathi, we're onto something real.

This micro-project will help us turn this idea into a proper startup proposal - with a solid business model, financial plan, and pitch deck ready to go.

2.0 **Literature Review** *(Students should provide the relevant knowledge/information and existing status, from journals/technical magazines, brochures, websites and other sources referred/required to complete the selected micro-project in about 200 to 500 words.)*

Global players like **Sense (USA)** and **Smappee (Europe)** make smart energy monitors, but they cost ₹15,000-₹30,000 and aren't designed for Indian appliances. Their algorithms don't recognize desert coolers, geysers, or mixer grinders.

What's happening in India?

Government schemes are actually pretty generous:

- **PM Surya Ghar Yojana** gives free electricity up to 300 units if you install solar (₹75,021 crore allocated)
- **UJALA Scheme** has distributed 36 crore LED bulbs across India, saving 47,000 million kWh annually

But here's the catch - a CEEW study found that **81% of people don't even know these schemes exist**. And among those who know, 73% find the application process too complicated.

Current Indian options:

Solution	Problem
Discom portals/apps	Only show monthly total, not real-time appliance breakdown
Smart plugs (Wipro, Philips)	Need one per appliance - expensive (₹1,200-2,500 each)
Solar companies	Focus only on installation, not ongoing monitoring
No integrated solution	Nothing that combines tracking + savings + subsidies

What our survey found (n=54):

We asked people what they actually want:

- 83% agreed they have no visibility into appliance consumption
- 52% don't track their electricity at all
- 44% said "appliance-wise breakdown" is their top priority
- 78% would use UrjaSaathi
- 54% are willing to pay ₹299-₹499 for it
- 61% prefer Hinglish/Hindi over pure English

The technology behind it:

There's a technology called **NILM (Non-Intrusive Load Monitoring)**. First proposed by Hart in 1992, it uses AI to figure out which appliance is running just by looking at the overall power signal. Recent research by Zhang et al. (2024) shows that modern deep learning models can achieve 92-95% accuracy. Studies from IIT Bombay (2023) specifically focused on Indian appliance signatures, proving it can work with our unique appliances.

The gap we're filling:

Technically, we know it's possible. The market need is proven. The gap is making it affordable, accurate for Indian homes, and user-friendly for regular families. That's exactly what UrjaSaathi aims to do.

3.0 Proposed Methodology *(Students should provide a detailed Flowchart/Procedure in brief (about 200 to 500) that will be followed to complete the micro-project.)*

We'll follow a simple 6-phase approach from now until April 2026:

PHASE 1: DATA ANALYSIS & INSIGHTS (Feb 18 - Feb 25)

- Analyze our 54 survey responses in detail
- Identify key patterns and customer pain points
- Create user personas (who are we building for?)

PHASE 2: SEGMENTATION & DESIGN (Feb 26 - Mar 8)

- Perform customer segmentation analysis
- Identify target segments (Tech-Savvy Savers, Budget Homeowners, etc.)
- Create wireframes on Figma (how the app looks and works)

PHASE 3: FEATURE & BUSINESS MODEL (Mar 9 - Mar 19)

- Prioritize features using MoSCoW method:
 - **Must have:** Appliance breakdown, bill prediction, Hinglish UI
 - **Should have:** AI savings tips
 - **Could have:** Solar subsidy assistant
 - **Won't have (yet):** Voice control
- Complete Business Model Canvas
- Finalize pricing: Free basic, ₹449 premium, ₹2,999 hardware kit

PHASE 4: FINANCIAL PLANNING (Mar 20 - Mar 28)

- Create 3-year financial projections (P&L, cash flow)
- Calculate break-even point
- Determine funding requirement (~₹12 lakhs)

PHASE 5: PITCH DECK DEVELOPMENT (Mar 29 - Apr 5)

- Build 10-slide investor presentation covering:
 1. Problem
 2. Solution
 3. Market Size
 4. Product
 5. Business Model
 6. Traction
 7. Competition
 8. Team
 9. Financials
 10. Ask

PHASE 6: FINALIZATION & SUBMISSION (Apr 6 - Apr 15)

- Internal review and refinement
- Faculty feedback incorporation
- Final project submission

4.0 Tentative Action Plan *(Sequence of activities and time (Gantt Chart) required to complete the project.)*

S.No	Activity	Start Date	End Date	Responsible Member
1	Course commenced	10/01/2026	–	–
2	Idea identification and team formation	15/01/2026	20/01/2026	ALL MEMBERS
3	Preliminary market research	22/01/2026	05/02/2026	ALL MEMBERS
4	Survey design and pilot testing	06/02/2026	12/02/2026	SANTHOSA PRIYAN K A
5	Data collection (54 responses)	13/02/2026	17/02/2026	KALAISELVAN R P
6	Survey data analysis and insights	18/02/2026	22/02/2026	NISTCHAL MARCHUS Y
7	User persona development	23/02/2026	25/02/2026	DHANUSH DHARMENDRA
8	Customer segmentation analysis	26/02/2026	02/03/2026	SANTHOSA PRIYAN K A
9	Wireframing and UI mockups	03/03/2026	08/03/2026	KALAISELVAN R P
10	Feature prioritization (MoSCoW)	09/03/2026	11/03/2026	NISTCHAL MARCHUS Y

5.0 Tentative Resources (if any) Required *(major resources such as raw materials, some machining facilities, software etc.)*

S.No	Item	Broad Specifications	Qty	Cost (₹)
1	Laptops for development	Windows 11, 8GB+ RAM (already owned)	4	₹0
2	Software tools	Figma (free), VS Code (free), React Native (open source)	–	₹0
3	Cloud services	Firebase / AWS Free Tier	–	₹0
4	Hardware prototype kit	ESP32, current sensors, CT clamps, breadboard	2	₹2,500
5	Internet connection	Broadband (100 Mbps)	2 months	₹2,000
6	Stationery & printing	Project reports, charts, binding	–	₹1,000
7	User interview incentives	Amazon vouchers for 10 participants	10	₹2,000
8	Domain registration	urjasaathi.in (optional)	1	₹999
	TOTAL ESTIMATED COST			₹8,499
